INCH-POUND
MIL-DTL-0048669A (AR)
29 March 2004
USED IN LIEU OF
MIL-C-48669(5) (AR)
1 April 1986

DETAIL SPECIFICATION

CARTRIDGE, 40MM - M550
ESCAPEMENT ASSEMBLY

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC). Other activities in the Department of Defense may use this interim revision or may continue using MIL-C-48669 (AR).

1. SCOPE

1.1 Scope. This specification covers the metal parts, assemblies and loading of escapement assembly for use in the M550 Fuze and the Cartridge, 40MM - M918 TP.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-14978 Detonator, Stab, M55 Loading, Assembling and Packing

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000, or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

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MIL-A-70625	Automated Acceptance Inspection Equipment Design, Testing and Approval, of
MIL-C-48683	Cartridge, 40MM: M918 TP; Projectile Assembly
MIL-F-50869	Fuze PIBD: M550, Less Spitback Assembly

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DoD Preferred Method for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

US ARMY (ARDEC) DRAWINGS (see 6.3)

8886357 - Escapement Assembly
8886358 - Housing, Escapement
8886359 - Plate, Top
8886360 - Fuze, PIBD: M550, Less Spitback
8886361 - Detent
8886362 - Spring, Detent
8886363 - Rotor Assembly
8886365 - Verge Assembly
8886366 - Verge
8886367 - Arbor, Verge
8886368 - Pinion Assembly
8886369 - Pinion
8886370 - Starwheel
9219871 - Rotor and Gear Assembly
9219872 - Sector, Gear
9219873 - Rotor
9219874 - Brushing
9219876 - Shaft, Rotor
9399374 - Cartridge, 40MM: M918 TP Projectile Assembly
12999562 - Pin, Setback
12999563 - Spring, Setback

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM E1255	Standard Practice for Radioscopy
ASTM D1474	Indentation Hardness of Organic Coatings, Test for

(Copies may be requested online at www.astm.org or ordered from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials, parts and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Arming. The escapement assembly shall arm when rotated at 2900 RPM.

3.4 Arming time. The escapement assembly arming time shall be 0.220 to 0.350 seconds when rotated at 3750 ± 50 RPM.

3.5 Non-arming. The escapement assembly shall not arm when rotated at 1500 RPM.

3.6 Setback pin retraction force and spring removal. The force required to push the setback pin through the setback spring of the escapement shall be 30 to 63 ounces.

3.7 Transportation vibration. The escapement as a component of the higher assemblies (M550 Fuze assembly Dwg. 8886360 and/or the M918 TP Projectile assembly (Dwg. 9399374) shall not become armed or partially armed, shall not have any safety features missing or not in its proper position, and shall meet the requirements of functioning (3.10) and non-functioning (3.11).

3.8 Jolt. The escapement as a component of the higher assemblies (M550 Fuze assembly Dwg. 8886360 and/or the M918 TP Projectile assembly (Dwg. 9399374) shall not become armed or partially armed and shall be safe for handling and disposal.

3.9 Jumble. The escapement as a component of the higher assemblies (M550 Fuze assembly Dwg. 8886360 and/or the M918 TP Projectile assembly (Dwg. 9399374) shall not become armed or partially armed and shall be safe for handling and disposal.

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3.10 Functioning. The escapement as a component of the higher assembly, the M550 Fuze (Dwg. 8886360) and/or the M918 TP Projectile assembly (Dwg. 9399374), shall meet all functioning requirements for that assembly.

3.11 Non-functioning. The escapement as a component of the higher assembly, the M550 Fuze (Dwg. 8886360) and/or the M918 TP Projectile assembly (Dwg. 9399374), shall meet all non-functioning requirements for that assembly.

3.12 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.13 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough and workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

4. VERIFICATION

TABLE I. Requirement/Inspection Cross Reference Matrix

Method of inspection
N/A – Not Applicable (N/A)
1 – Analysis
2 – Demonstration
3 – Examination
4 – Test

Classes of inspection
A – First article
B – Conformance

SECTION 3 REQUIREMENT	INSPECTION METHODS					INSPECTION CLASS		SECTION 4 INSPECTION METHOD
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.4
3.3					X	X	X	4.5.11
3.4					X	X	X	4.5.12
3.5					X	X	X	4.5.13
3.6					X	X	X	4.5.15
3.7					X	X	X	4.5.14
3.8					X	X	X	4.5.16
3.9					X	X	X	4.5.17
3.10					X	X	X	4.5.18
3.11					X	X	X	4.5.19
3.12				X		X	X	4.5.21
3.13				X		X	X	4.5.22

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (4.3)
b. Conformance inspection (4.4)

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4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First article inspection. When specified in the solicitation or contract, a sample of the escapement shall be subjected to first article inspection.

4.3.1 First article quantity. The contractor shall submit a first article for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawings	Quantity
Housing, Escapement	8886358	25
Plate, Top	8886359	25
Detent	8886361	25
Spring, Detent	8886362	25
Verge	8886366	25
Arbor, Verge	8886367	25
Pinion	8886369	25
Starwheel	8886370	25
Sector, Gear	9219872	25
Rotor	9219873	25
Bushing	9219874	25
Shaft, Rotor	9219876	25
Pin, Setback	12999562	125
Spring, Setback	12999563	125
Verge Assembly	8886365	125
Pinion Assembly	8886368	125
Rotor and Gear Sector Assembly	9219871	125
Escapement Assembly	8886357	125

4.3.2 Inspections to be performed. The first article inspection shall be 100% examination for defects of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and the tests specified in Table II herein.

4.3.3 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE II. First Article Inspection

Escapement Assembly and Components

EXAMINATION OR TEST	NO. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Spring, Setback (Dwg. 12999563)			
Push Test of simulated setback pin	25	3.2	4.5.2
Verification of heat treat	1/	3.2	4.5.3
Verge Assembly (Dwg. 8886365)			
Torque Test	50	3.2	4.5.6
Disassembly test	50	3.2	4.5.7
Pinion Assembly (Dwg. 8886368)			
Push off test	50	3.2	4.5.8
Torque Test	50	3.2	4.5.9
Rotor and Gear Sector Assembly (Dwg. 9219871)			
Static push test	50	3.2	4.5.10
Escapement Assembly (Dwg. 8886357)			
Push off test	125 2/	3.2	4.5.4
Arming test	125 2/	3.3	AAIE/4.5.1, 4.5.11
Arming time, min	125 2/	3.4	AAIE/4.5.1, 4.5.12
Non arming	125 2/	3.5	AAIE/4.5.1, 4.5.13
Transportation vibration	3/	3.7	4.5.14, 4.5.20
Pin Retraction and Spring Removal Test	100	3.6	4.5.15
Jolt	3/	3.8	4.5.16
Jumble	3/	3.9	4.5.17
Functioning	3/	3.10	4.5.18
Non-functioning	3/	3.11	4.5.19

NOTES:

1/ Three (3) setback springs having the lowest push force from the simulated setback pin push test.

2/ The quantity to be tested is determined by total required for the tests defined herein for FAT plus the higher assembly FAT requirement quantities (i.e. Jolt, Jumble, Functioning, Non-functioning). All escapements required to support higher assembly FAT requirements shall be subjected to these tests. All escapements shall be tested 100%.

3/ This requirement is tested at a higher assembly level. See either MIL-DTL-48683 (M918 TP; Projectile Assembly) or MIL-DTL-50869 (M550 Fuze Less Spitback Assembly) for quantities.

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot of escapements shall be produced with components complying with the following constraints:

a. Detonators from one interfix number, from one manufacturer.
b. Metal parts or components from one interfix number, from one manufacturer.

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4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are as defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.7. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Pin, Setback	SHEET 1 OF 1	12999562
			NEXT HIGHER ASSEMBLY 8886357

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Length below flange, min.	100%	3.2	AAIE/4.5.1
MAJOR				
101.	Height of pin head	Level IV	3.2	Gage
MINOR				
201.	Evidence of poor workmanship	Level II	3.13	Visual/4.5.22
NOTES:				

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.2	Spring, Setback	SHEET 1 OF 1	12999563
			NEXT HIGHER ASSEMBLY 8886357

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None defined			
MAJOR				
101.	Diameter between tangs	Level IV	3.2	Gage
102.	Tang broken, bent or missing	Level IV	3.2	Visual with magnification
103.	Push test of simulated setback pin	100 2/	3.2	4.5.2
104.	Verification of heat treat	3 Samples 1/	3.2	4.5.3
MINOR				
201.	Length, max	Level II	3.2	Gage
202.	Evidence of poor workmanship	Level II	3.13	Visual/4.5.22

NOTES:
1/ Per heat treat batch.
2/ Accept on zero (0) and reject on one (1).

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 8886365
4.4.2.3	Verge Assembly			NEXT HIGHEST ASSEMBLY 8886357
CLASSIFICATION	**EXAMINATION OF TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE / INSPECTION METHOD**
CRITICAL	None defined			
MAJOR				
101.	Bearing diameters (see Dwg. 8886367)	Level IV	3.2	Gage
102.	Thickness of verge (see Dwg. 8886366)	Level IV	3.2	Gage
103.	Outside diameter of verge (see Dwg 8886366)	Level IV	3.2	Gage
104.	Depth of cutout on verge (see Dwg. 8886366)	Level IV	3.2	Gage
105.	Distance from shoulder of arbor to end of bushing	Level IV	3.2	Gage
106.	Perpendicularity of verge with bearing diameter	Level IV	3.2	Gage
107.	Torque test	Level III	3.2	4.5.6
108.	Disassembly test	Level III	3.2	4.5.7
MINOR				
201.	Lubricant missing	Level II	3.2	Visual
202.	Evidence of poor workmanship	Level II	3.13	Visual/4.5.22
NOTES:				

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF1	DRAWING NUMBER 8886368
4.4.2.4	Pinion Assembly		NEXT HIGHER ASSEMBLY 8886357

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None defined			
MAJOR				
101.	Bearing diameter (2) (See Dwg. 8886369)	Level IV	3.2	Gage
102.	Starwheel thickness (See Dwg. 8886370)	Level IV	3.2	Gage
103.	True position of outside diameter of starwheel with bearing diameters	Level IV	3.2	Gage
104.	Distance from top of pinion to top of bushing	Level IV	3.2	Gage
105.	Perpendicularity of starwheel with bearing diameters	Level IV	3.2	Gage
106.	Push off test	Level III	3.2	4.5.8
107.	Torque test	Level III	3.2	4.5.9
MINOR				
201.	Lubricant missing	Level II	3.2	Visual
202.	Evidence of poor workmanship	Level II	3.13	Visual/4.5.22

NOTES:

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.5	Rotor and Gear Sector Assembly	SHEET 1 OF 1	9219871
			NEXT HIGHER ASSEMBLY 8886363

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None defined			
MAJOR				
101.	Static push off test	32 1/	3.2	4.5.10
102.	Parallelism of rotor to gear sector	Level IV	3.2	Gage
MINOR				
201.	Evidence of poor workmanship	Level II	3.13	Visual/4.5.22

NOTES:
1/ Accept on zero (0) and reject on one (1).

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.6	Rotor Assembly 1/	SHEET 1 OF 1	8886363
			NEXT HIGHEST ASSEMBLY 8886357

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None defined			
MAJOR				
101.	Detonator stake missing (2) min. required	Level IV	3.2	Visual
102.	Detonator inverted	Level IV	3.2	Visual
103.	Detonator above flush	Level IV	3.2	Gage/Probe
104.	Discernible bulge on detonator	Level IV	3.2	Visual
105.	Push off test	Level III	3.2	4.5.4
106.	Check test for deterioration	As Required	3.2	4.5.5
MINOR				
201.	Evidence of poor workmanship	Level II	3.13	Visual/4.5.22

NOTES:
1/ All requirements for this assembly may be checked as part of the final escapement assembly less Setback pin and spring.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.7	Escapement Assembly	SHEET 1 OF 1	8886357
			NEXT HIGHER ASSEMBLY ----------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Arming time, min	100%	3.4	AAIE/4.5.1, 4.5.12
2.	Rotor in armed or partially armed position 1/	100%	3.2	AAIE/4.5.1
3.	Non-arming	100%	3.5	AAIE/4.5.1, 4.5.13
4.	Detent spring missing	100%	3.2	AAIE/4.5.1
5.	Detent missing	100%	3.2	AAIE/4.5.1
6.	Setback spring missing	100%	3.2	AAIE/4.5.1
7.	Setback pin missing or retracted 2/	100%	3.2	AAIE/4.5.1
8.	Pinion or verge assembly missing	100%	3.2	AAIE/4.5.1
9.	Transportation vibration	5/	3.7	4.5.14, 4.5.20
10.	Jolt	5/	3.8	4.5.16, 4.5.20
11.	Jumble	5/	3.9	4.5.17, 4.5.20
12.	Detent not engaged in teeth of sector gear 3/	100%	3.2	AAIE/4.5.1
13.	Setback pin not blocking rotor 4/	100%	3.2	AAIE/4.5.1
14.	Non-Functioning	5/	3.11	4.5.19
MAJOR				
101.	Arming test	100%	3.3	4.5.11
102.	Functioning	5/	3.10	4.5.18
103.	Pin retraction and spring removal test	100 6/	3.6	4.5.15
104.	Improper maximum height at 3 places (120 degrees apart) from base of escapement housing to top plate	Level IV	3.2	Gage
MINOR				
201.	Face of posts above flush after stake	Level II	3.2	Gage
202.	Evidence of poor workmanship	Level II	3.13	Visual/4.5.22

NOTES:

1/ See 6.4 for definition of armed and partially armed.

2/ A retracted setback pin is a setback pin in an escapement assembly with the dimension (excluding cut off nib) from the top of it to the bottom of the escapement housing, Dwg. 8886358, being 0.040 or greater.

3/ Either the protrusion of the detent into the sector gear teeth or the seating of the detent in its forward position shall be inspected.

4/ The setback pin shall be checked to be blocking the rotor in accordance with Dwg. 8886357.

5/ This requirement is tested at higher assembly. See MIL-DTL-48683 (M918 TP; Projectile Assembly) or MIL-DTL-50869 (M550 Fuze Less Spitback Assembly) for quantities.

6/ Accept on zero (0) and reject on one (1).

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4.5 Methods of inspection.

4.5.1 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE) prove-out shall be performed in accordance with MIL-A-70625 (see 6.6).

4.5.2 Push test of simulated setback pin into spring setback. The setback spring shall be placed in an approved fixture and the force necessary to push the simulated setback pin through the tangs shall be measured. If the measured force for any spring in the sample is less than the minimum allowed by the applicable drawing the spring lot shall be rejected and be set aside for disposition in accordance with MIL-STD-1916. If the measured force for any two or more springs is more than the maximum allowed by the applicable drawing the spring lot shall be rejected and be set aside for disposition in accordance with MIL-STD-1916. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.3 Verification of heat treat of spring setback. The setback spring shall be subjected to the Knoop hardness reading in accordance with ASTM D1474. After completion of the push test of simulated setback pin, the three (3) setback springs exhibiting the lowest push test force shall be subjected to this test. If any setback spring fails to comply with the requirement specified on the applicable drawing, the heat treatment batch represented by the sample shall be rejected and be set aside for disposition in accordance with MIL-STD-1916. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.4 Push off test of rotor assembly. The rotor assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied to the detonator, as a minimum. The push off force shall not exceed 10% of the specified minimum value. The push off force shall be applied uniformly over the surface of the detonator. This test shall be conducted prior to tests specified in 4.5.11, 4.5.12, and 4.5.13 when done as part of the escapement assembly. This test is considered a non-destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.5 Check test for deterioration of detonators. If the total time between original acceptance of any lot and the assembly of that lot into the escapement assembly exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since the previous tests, the detonator lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-D-14978 immediately before the detonator lot is assembled into the escapement assembly. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the escapement assembly.

4.5.6 Torque test of verge assembly. The verge assembly shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied, as a minimum. The torque applied to the verge assembly shall not exceed 20% of the specified minimum value. The bearing surfaces of the verge assembly shall not be utilized for retention of the assembly

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during the torque test if the parts are intended to be returned to the lot. This test is considered a non-destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.7 Disassembly test of verge assembly. The verge assembly shall be placed in an approved fixture and the disassembly force specified on the applicable drawing shall be applied, as a minimum. The disassembly force shall not exceed 10% of the specified minimum value. The disassembly force shall be applied uniformly over the surface of the verge. This test is considered a non-destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.8 Push off test of pinion assembly. The pinion assembly shall be placed in an approved fixture and the push off force specified on the applicable drawing shall be applied, as a minimum. The push off force shall not exceed 10% of the specified minimum value. The push off force shall be applied uniformly over the surface of the star wheel. This test is a considered non-destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.9 Torque test of pinion assembly. The pinion assembly shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied, as a minimum. The torque applied to the pinion assembly shall not exceed 20% of the specified minimum value. The bearing surfaces of the pinion assembly shall not be utilized for retention of the assembly during the torque test if the parts are intended to be returned to the lot. This test is considered a non-destructive test within the specified limits. Parts so tested may be returned to the lot.

4.5.10 Static push test of rotor and gear sector assembly. The rotor and gear sector assembly shall be placed in an approved fixture and the push off force specified on the applicable drawing shall be applied, as a minimum. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.11 Arming test. The escapement assembly shall be placed in an approved fixture and rotated up to the maximum arming speed specified in 3.3. This test is a non-destructive test. Parts successfully tested may be returned to the lot after the rotor has been returned to the safe position.

4.5.12 Arming time. The escapement assembly shall be placed in an approved fixture and rotated at the speed specified in 3.4. The arming time shall be measured. This test is a non-destructive test. Parts successfully tested shall be returned to the lot after the rotor has been returned to the safe position. Parts failing to achieve the minimum arming time of the applicable drawing shall be rejected as a critical defect and immediately removed and isolated from the lot.

4.5.13 Non-arming. This test shall be performed after completion of the tests specified in 4.5.11 and 4.5.12. The escapement assembly shall be placed in an approved fixture and rotated at the non-arming speed specified in 3.5, as a minimum. This test is a non-destructive test. Parts successfully tested may be returned to the lot. Parts failing this test by arming shall be rejected as a critical defect and immediately removed and isolated from the lot.

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4.5.14 Transportation vibration. This test shall be in accordance with the requirements of the higher assembly specification MIL-DTL-48683 (M918 TP; Projectile Assembly) or MIL-DTL-50869 (M550 Fuze Less Spitback Assembly).

4.5.15 Pin Retraction and Spring Removal Test. The escapement assembly shall be placed in an approved fixture and the retraction force required to push the setback pin through the tangs shall be measured. If the measured retraction force for any escapement in the sample is less than the minimum allowed by the applicable drawing the escapement lot shall be rejected and be set aside for disposition in accordance with MIL-STD-1916. If the measured retraction force for any two or more escapements is more than the maximum allowed by the applicable drawing the escapement lot shall be rejected and be set aside for disposition in accordance with MIL-STD-1916. After the setback pin retraction force has been measured the escapement shall be placed in an approved fixture and the removal force required to pull the spring from the housing shall be measured. The spring shall be removed by grabbing one or more of the tangs. If the measured spring removal force of two or more escapements is less than the minimum allowed by the applicable drawing the escapement lot shall be rejected and be set aside for disposition in accordance with MIL-STD-1916. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.16 Jolt. This test shall be in accordance with the requirements of the higher assembly specification MIL-DTL-48683 (M918 TP; Projectile Assembly) or MIL-DTL-50869 (M550 Fuze Less Spitback Assembly).

4.5.17 Jumble. The test shall be in accordance with the requirements of the higher assembly specification MIL-DTL-48683 (M918 TP; Projectile Assembly) or MIL-DTL-50869 (M550 Fuze Less Spitback Assembly).

4.5.18 Functioning. The test shall be in accordance with the requirements of the higher assembly specification MIL-DTL-48683 (M918 TP; Projectile Assembly) or MIL-DTL-50869 (M550 Fuze Less Spitback Assembly).

4.5.19 Non-functioning. The test shall be in accordance with the requirements of the higher assembly specification MIL-DTL-48683 (M918 TP; Projectile Assembly) or MIL-DTL-50869 (M550 Fuze Less Spitback Assembly).

4.5.20 X-ray of escapement assembly. Examination shall be performed with X-ray equipment or any other method acceptable to the contracting officer. Radioscopic examination equipment, procedures, and personnel shall be in accordance with the requirements of ASTM E1255. The escapement assembly shall be examined for compliance with 3.2. The test shall be conducted in two (2) planes:

a. From the top of the escapement assembly down.
b. From the side of the escapement assembly.

4.5.21 Lot identification. Visually verify that each lot of escapements is identified as the type, caliber, model and lot number in accordance with MIL-STD-1168.

4.5.22 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.13.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The assemblies covered by this specification are intended for use with the Cartridge, 40MM - M918 TP and the M550 Fuze. Additional requirements may exist beyond those identified here for those individual applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number (w/revision level) and date of all reference documents.
c. ASSIST Online to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1).
d. Requirements for submission of first article sample.
e. Requirement for submission of design verification sample
f. Certificate of conformance for each lot or shipment of product.
g. Packaging requirements, if other than specified in Section 5.
h. Quality Conformance Inspection, if other than specified in Section 4 of this specification.
i. Provisions for critical characteristic controls.
j. Requirement for submission of inspection equipment designs.
k. Applicable National Stock number (NSN)
l. Serialization requirements, if applicable.
m. Government Furnished Material or Equipment.
n. Lists of drawings, publications and specifications, showing applicable revision dates.
o. Critical inspection equipment requirement (see 6.6).

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.4 Definitions.

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6.4.1 Armed. The detonator is centered in the spitback hole.

6.4.2 Partially Armed. A rotor assembly with two or more teeth in the armed position.

6.4.3 Automated acceptance inspection equipment. Equipment for which no, or minimal, human involvement is required in the acceptance determination. This would include equipment employing probes/sensors/transducers that are automatically manipulated to perform measuring and detection functions. If readout is provided, the equipment will be categorized as automated even if operator interpretation is needed in the accept/reject decision.

6.5 Key Word Listing.

M918 Projectile
M433 Cartridge
M550 Fuze

6.6 Critical Inspection Equipment.

6.6.1 System Reliability. The contractor should provide for each defect identified as Critical, a manufacturing and inspection system that assured no more than one in a million contain the defect. This should be assured by controlling the maximum defect rate produced and the error rate of the inspection equipment such that the product of the two terms is less than one in a million.

6.6.2 Maximum defect rate produced. The maximum defect rate produced should be defined as the largest defect rate expected for the characteristics of concern on a monthly or lot basis. The contractor should establish the maximum defect rate, however, it may not exceed one percent without approval of the procuring contracting officer. Once established, the contractor should monitor the defect rate to assure it does not exceed the maximum rate allotted. If the established maximum defect rate is exceeded, the production of that feature should be terminated until the cause is determined and corrected. Additionally, all products for that period of time should be rejected and reinspected.

6.6.3 Maximum error rate of the inspection system. The error rate of the inspection equipment should be defined as the expected ratio of the number of defective parts accepted to the number of defective parts inspected by the equipment. The contractor, to meet the system reliability as defined in contract, defines the maximum error rate allowed. However, it may not exceed 1/500 without approval of the procuring contracting officer. Based on the maximum error rate defined for the equipment, the contractor should develop demonstration test procedure to demonstrate the error rate of the equipment. The test should be performed using defective parts or reject standards. No part or standard should be accepted during the test. If a part or standard is accepted, the cause for failure should be isolated and corrected and the test rerun. Unless otherwise specified in the detailed item specification, the minimum number of test samples to be run should be equal to 0.7 divided by the error rate (e.g. If the required error rate is 1/1000, the sample size would be 0.7 divided by (1/1000) = 700.)

6.6.4 Periodic verification. Periodic verification of the system error rate should be performed.

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6.6.5 Redundant inspection equipment. The contractor may elect to use redundant inspection equipment in lieu of using a single piece of equipment with a very low error rate. Where redundant equipment is used, the inspection system error rate is defined as the error rates of the multiple pieces of equipment multiplied together.

6.6 Revision/Amendments History. The following is a record of changes incorporated into this document

ECP	Description	Date Approved
R2S3021	Add a hole to top plate per attached sketch	09/25/2002
R2S3018	Change test quantity identified in "Pin Retraction and Spring Removal Test" from 80 to 100 in paragraphs defining the test.	07/31/2002
R2S2021	• Implementation of new setback pin and spring for the M550 escapement used in the M550 fuze and M918 TP. • Clarification of the minimum arming time requirements in the item specifications.	07/01/2002
R1Q2005	Add an additional setback pin inspection.	03/06/2001
H0S2001	• Add setback spring height from the bottom of escapement housing • Add overall height of escapement housing	10/05/2000
R04Q2022	Initial release of M550 escapement MIL-DTL specification	Aug 2004

6.7 Interim Revision This interim revision has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim revision has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Army-AR
(Project 1390-2006-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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